

YORKTON, SK, Canada

WMO: 712920

Lat: 51.2601N

Lon: 102.4615W

Elev: 1635

StdP: 13.85

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.6	-22.0	-35.2	0.8	-26.4	-30.5	1.1	-21.7	29.0	15.0	25.6	11.4	7.3	300	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	84.8	67.1	81.4	65.5	78.4	63.5	70.2	80.3	67.8	78.1	65.8	75.1	11.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	103.9	75.8	64.2	95.4	73.3	62.2	88.8	70.6	35.3	80.3	33.2	77.9	31.5	75.0	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.8	19.5	DB	-33.6	91.0	5.6	2.9	-37.6	93.1	-40.9	94.8	-44.1	96.4	-48.1	98.5	
			WB	-33.8	74.7	5.5	3.0	-37.8	76.9	-41.1	78.7	-44.2	80.4	-48.2	82.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	3.5	6.9	19.3	37.2	50.6	60.3	65.2	63.1	54.1	39.6	22.2	8.5
	DBStd	24.65	15.56	13.99	14.00	11.11	8.72	6.46	5.46	6.15	8.04	9.08	12.41	14.25
	HDD50	6613	1440	1206	952	401	101	5	0	1	48	339	833	1287
	HDD65	10735	1905	1626	1417	835	452	168	66	108	336	787	1283	1752
	CDD50	1514	0	0	0	15	119	313	472	408	170	17	0	0
	CDD65	160	0	0	0	0	4	25	72	51	8	0	0	0
	CDH74	1871	0	0	0	5	109	296	697	596	163	5	0	0
	CDH80	431	0	0	0	0	19	58	157	164	33	0	0	0
	WSAvg	9.5	9.8	9.7	9.9	10.2	10.6	9.2	8.3	8.4	9.3	10.0	9.8	9.5
	PrecAvg	18.0	0.9	0.7	1.1	0.9	2.1	2.8	2.5	2.3	2.0	1.0	0.8	0.9
Precipitation	PrecMax	23.3	2.2	1.9	2.3	3.2	4.5	6.0	5.6	5.6	4.3	3.0	2.1	1.6
	PrecMin	10.7	0.0	0.2	0.2	0.0	0.2	0.4	0.4	0.6	0.1	0.1	0.0	0.3
	PrecStd	3.4	0.6	0.4	0.5	0.6	1.1	1.5	1.4	1.5	1.1	0.8	0.5	0.4
	PrecStd	3.4	0.6	0.4	0.5	0.6	1.1	1.5	1.4	1.5	1.1	0.8	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	38.8	52.5	72.3	82.6	85.8	88.6	89.7	84.4	72.2	53.9	40.5
		MCWB	35.0	35.0	43.6	53.5	59.7	68.7	70.7	68.1	64.9	56.4	45.4	35.5
	2%	DB	34.6	34.1	45.0	65.6	76.9	80.9	84.3	84.5	78.9	64.2	46.4	35.8
		MCWB	31.9	31.8	39.3	50.2	56.9	64.3	69.1	66.9	61.2	50.9	40.0	32.8
	5%	DB	30.4	30.8	40.1	60.2	71.9	77.2	80.9	80.4	73.8	58.5	41.4	31.4
		MCWB	28.4	28.9	36.1	46.8	54.7	62.1	66.8	65.3	59.4	48.5	36.7	29.1
	10%	DB	26.1	27.2	36.5	54.6	67.6	73.7	77.8	76.7	69.0	53.6	37.5	27.5
		MCWB	24.6	25.6	33.5	44.3	52.3	60.2	65.3	63.3	56.9	46.2	34.4	25.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	35.2	44.7	54.7	62.4	72.5	73.9	71.7	67.2	57.3	46.1	35.8
		MCDB	38.1	37.6	51.4	70.0	74.7	81.2	83.2	83.0	79.3	69.1	53.1	40.2
	2%	WB	31.9	32.1	39.3	51.0	59.0	66.8	71.1	69.0	63.7	52.9	40.6	32.8
		MCDB	34.4	34.0	44.6	63.1	72.3	76.5	80.9	80.5	74.6	61.4	45.3	35.8
	5%	WB	28.4	29.0	36.2	48.0	56.2	64.1	68.7	66.9	60.9	49.4	37.2	29.3
		MCDB	30.3	30.7	40.0	58.2	69.5	73.4	78.2	77.4	71.2	57.3	40.5	31.3
	10%	WB	24.6	25.6	33.7	45.1	53.7	61.8	66.5	64.7	57.8	46.5	34.4	26.0
		MCDB	26.2	27.2	36.3	54.1	65.0	70.8	75.6	74.2	67.4	53.1	37.4	27.5
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.7	17.0	19.8	22.7	19.7	20.4	21.9	21.4	17.8	14.6	14.8
		MCDBR	17.7	17.2	18.6	28.7	29.4	25.5	25.1	28.1	29.7	27.5	20.1	17.9
		MCWBR	16.0	15.2	13.9	15.9	13.9	12.2	12.7	13.6	15.5	16.3	14.4	15.4
	5% WB	MCDBR	17.8	16.2	18.1	26.4	26.1	21.7	22.8	24.7	26.9	26.2	18.8	17.4
		MCWBR	16.3	14.4	13.7	15.6	13.5	11.8	13.0	13.2	15.2	15.9	14.0	15.2
Clear-Sky Solar Irradiance	taub		0.246	0.238	0.281	0.318	0.325	0.348	0.364	0.370	0.328	0.280	0.248	0.234
	taud		2.288	2.352	2.375	2.435	2.469	2.431	2.400	2.369	2.469	2.565	2.485	2.346
	Ebn at Noon		251	291	294	294	295	288	280	272	274	273	255	240
	Edh at Noon		20	25	30	32	33	35	35	34	28	21	17	16
All-Sky Solar Radiation	RadAvg		398	748	1200	1566	1838	1879	1989	1611	1192	696	363	279
	RadStd		31	62	98	112	160	148	110	132	104	92	48	26

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-0.97	N/A	N/A	N/A	N/A

Nomenclature: See separate page